Time Series Forecasting with Clustering

1.

The dataset contains daily energy consumption for 17,547 Austrian households. 2023 is the training year (365 days). 2024 is the test year (366 days, leap year). The goal is to predict every household's daily consumption for all 366 days of 2024.

The approach has two parts. First, cluster the 2023 households into groups based on how their consumption behaves - seasonal shape, weekly rhythm, scale. Second, train one forecasting model per cluster so each model only sees households that behave similarly, which produces better predictions than a single model on all 17,547 households at once.

As a baseline, one global model is also trained on all households combined. Comparing cluster model MAE versus global model MAE is the headline result of the project.

2. Folder structure

```
RDK Project (3)/
├── Data/
│   ├── sample_23.csv          training data  (17547 × 365)
│   └── sample_24.csv          test data      (17547 × 366)  - holdout only
├── notebooks/
│   ├── 01_EDA.ipynb
│   ├── 02_Preprocessing_Features.ipynb
│   ├── 03_Clustering_Experiments.ipynb
│   ├── 04_Forecasting.ipynb
│   └── 05_Probabilistic_Forecasting.ipynb
├── src/
│   ├── utils/config.py        all constants (RANDOM_SEED, FINAL_K, paths…)
│   ├── utils/data_loader.py   load_train_data() / load_test_data()
│   ├── preprocessing/         data cleaning + feature engineering classes
│   └── clustering/            HouseholdClusterer + ClusteringEvaluator
└── results/
    ├── diagnostics/           cluster assignment files
    ├── figures/               all saved plots (.png)
    ├── metrics/               MAE / RMSE / R² CSVs
    ├── models/                trained LightGBM .pkl files
    │   └── probabilistic/     quantile model .pkl files
    └── predictions/           prediction CSVs
```

3. Data files

sample_23.csv

Wide format - each row is one household, each column is a date (2023-01-01 through 2023-12-31). The first column is the household ID. Values are daily kWh consumption. Shape: 17,547 rows × 366 columns (1 ID + 365 dates). Mean daily consumption across all households is 9.13 kWh. Range is 0.0 to 188 kWh per household daily mean. Used exclusively for training and clustering.

sample_24.csv

Identical structure but covers 2024 (366 days). The same 17,547 household IDs appear in both files. This file is the ground truth for evaluation only - it is loaded once at the end to compute prediction errors. Nothing from this file is used during feature engineering, clustering, or model training.

4. Notebooks

01_EDA.ipynb - Exploratory Data Analysis

Loads sample_23.csv and examines it before any modelling. Does not produce files needed by downstream notebooks - it is purely analytical. Produces the most visually useful plots for presentations.

What it does:

- Computes per-household statistics: mean, std, CV, zero days, zero percentage, weekly autocorrelation, monthly profile variability.
- Splits households into 16,494 regular and 1,053 degenerate using phase-1 diagnostics.
- Runs outlier detection with both IQR (threshold 1.5) and Z-score (threshold 3.0) methods.
- Aggregates consumption by day-of-week (Mon–Sun averages) and by month (Jan–Dec averages) across all households.
- Computes z-normalized yearly profiles - removes scale, shows only shape - which directly motivated the shape-weighted feature design.
- Assigns heuristic EDA labels (low/medium/high consumer etc.). These are for understanding only, not used in clustering.

Key plots generated (all inline, not saved to disk):

Plot	What it shows
6-panel consumption histogram	Distribution of mean, std, CV, zero_days, min, max per household. Shows why households can't share one model.
15-panel sample time series	Raw daily curves for 15 random households. Visually proves behavioral diversity.
Monthly bar chart	Average monthly consumption across all households. Shows population-wide winter peak.
12-panel z-normalized profiles	Shape-only yearly curves. Reveals winter-peak, summer-peak, and flat profiles as distinct groups.
Heuristic category counts	Preliminary grouping for EDA context only.

02_Preprocessing_Features.ipynb - Feature Engineering

Cleans the raw data, identifies and removes degenerate households, and engineers the 32 features used for clustering. All outputs are saved to results/.

What it does:

- Runs DataCleaner.prepare_for_feature_clustering() to separate 16,494 regular households from 1,053 degenerate ones. Three degenerate types: all-zero (154 HH), near-constant (398 HH),

zero-heavy (398 HH). Outliers among regular households are kept because spikes are real behaviour.

- Runs FeatureEngineer.extract_main_clustering_features() to produce a (16494, 32) feature matrix across 5 blocks:
  - Level features (mean, std, CV, IQR, percentile ratios) - weight 1.0×
  - Weekly pattern (Mon–Sun averages, weekday vs weekend ratio) - weight 1.5×
  - Annual/seasonal (12 monthly averages, winter/summer contrast) - weight 2.0×
  - Shape-normalized (z-normalized monthly and weekly profiles) - weight 3.25×
  - Persistence/trend (ACF lag-7, rolling std, normalized slope) - weight 1.0×
- Runs FeatureSelector: variance filtering → correlation filtering (removes pairs > 0.90 correlated) → block weighting → StandardScaler normalization.
- PCA analysis available but disabled for the actual clustering run (FEATURE_USE_PCA = False in config.py).

Files saved:

File	Contents
results/method1_main_features.csv	Raw 32 features per household (16,494 × 32). Human-interpretable.
results/method1_filtered_features.csv	After variance + correlation filtering. Fewer columns.
results/method1_weighted_features.csv	After block weights applied. Shape features are at 3.25× scale.
results/method1_clustering_input.csv	Final StandardScaler-normalized input - what the clustering algorithm actually sees.
results/method1_feature_selector.pkl	Serialized FeatureSelector object including scaler, PCA model, and selected feature list.
results/method1_pca_loadings.csv	PCA component loadings (for inspection; PCA not used in clustering).

03_Clustering_Experiments.ipynb - Clustering

Determines the best number of clusters and produces the final frozen cluster assignments. All downstream notebooks depend on the files this saves.

What it does:

- Runs phase-1 diagnostics to confirm the degenerate household separation.
- Tests three clustering methods: Feature-Ward (hierarchical Ward linkage on engineered features), k-Shape (shape-based), and DTW-medoid (dynamic time warping).
- Evaluates k = 2 through 8 for each method using silhouette score, Davies-Bouldin index, Calinski-Harabasz index, cluster balance ratio, and a composite score.
- Composite score recommended k=3 for Feature-Ward (score 0.569). Final k overridden to 4 by manual inspection - at k=3, summer-peaking households merge into cluster 2 and lose their distinct identity. At k=4 they get their own cluster and their R^2 improves to 0.885.
- Runs HouseholdClusterer.prepare_clustering_handoff() which freezes the k=4 Feature-Ward result and saves the four output files.

Final cluster definitions:

Label	Name	Households	Character
0	moderate_winter_leaning	6,359	Mild winter peak, strong weekday routine. Most common type (38.6% of regular HH).
1	strong_winter_peaking_high_routine	2,757	Dec/Jan 2.5× higher than Jul/Aug. Very rigid weekday pattern. Likely electric heating.
2	lower_flatter_weekend_lift	5,807	Near-flat seasonality, unusually higher on weekends. Likely home-based or leisure users.
3	summer_peaking	1,571	Inverted profile, peaks Jun–Aug. Air conditioning or tourism-related properties.
-1	all_zero	154	Zero consumption all 365 days.
-2	near_constant	398	Extremely flat, std < 0.5 kWh. No seasonal or weekly signal.
-3	zero_heavy	501	More than 30% zero days with irregular spikes. Erratic.

Files saved:

File	Contents
results/diagnostics/cluster_assignments.csv	Maps every household_id to its cluster label. Index = household_id, column = cluster. This is the most critical output of the entire clustering phase.
results/diagnostics/cluster_sizes.csv	Number of households per cluster label. 4 rows × 3 columns.

File	Contents
results/diagnostics/cluster_monthly_profiles.csv	Mean monthly consumption profile per cluster. 4 rows × 13 columns (cluster + 12 months).
results/diagnostics/cluster_weekly_profiles.csv	Mean day-of-week profile per cluster. 4 rows × 8 columns (cluster + 7 days).

04_Forecasting.ipynb - Point Forecasting

Trains one LightGBM model per cluster plus one global baseline, generates 366-day predictions for all 17,547 households, evaluates performance, and saves all outputs.

What it does:

- Loads cluster assignments from results/diagnostics/. Splits into regular (16,494 HH) and degenerate (1,053 HH).
- Converts wide format to long format (one row per household-day). Stacks 2023 + 2024 before computing lag features so January 2024 lags can look back into December 2023.
- Builds 16 features per row: lag_1, lag_7, lag_14, lag_30, rolling_mean_7/14/30, rolling_std_7/14/30, day_of_week, month, day_of_year, is_weekend, week_of_year, household_mean.
- Trains 4 cluster LightGBM models on 2023 data (objective = regression_l1 to directly optimise MAE). Training sizes: C0 = 2.13M rows, C1 = 923K, C2 = 1.95M, C3 = 526K.
- Trains 1 global LightGBM model on all 5.53M regular-household training rows.
- Degenerate households: each gets its own 2023 mean predicted for all 366 days.
- Generates predictions for all 17,547 households × 366 days = 6,422,202 rows.
- Evaluates MAE, RMSE, MAPE, R^2 overall and per cluster. Saves feature importance, actual vs predicted plots, residual histograms.

LightGBM hyperparameters:

Parameter	Value
objective	regression_l1 (optimises MAE directly)
n_estimators	500
learning_rate	0.05
num_leaves	63
subsample / colsample_bytree	0.8 / 0.8
reg_alpha / reg_lambda	0.1 / 0.1
random_state	42

Results:

Cluster	Name	N HH	MAE	RMSE	R^2
-3	zero_heavy	501	4.404	11.709	0.257
-2	near_constant	398	0.335	1.167	0.356
-1	all_zero	154	0.080	1.479	-0.003
0	moderate_winter_leaning	6,359	1.953	4.334	0.815
1	strong_winter_peaking_high_routine	2,757	3.149	5.909	0.884
2	lower_flatter_weekend_lift	5,807	1.954	4.504	0.776
3	summer_peaking	1,571	2.145	4.549	0.885
ALL	overall	17,547	2.175	4.982	0.828

Files saved:

File	Contents
results/predictions/predictions_2024.csv	6,422,202 rows. Columns: household_id, date, predicted, actual, cluster, cluster_name, model_used.
results/metrics/forecast_metrics_overall.csv	Single row: overall MAE = 2.1749, RMSE = 4.9824, MAPE = 188.9, R^2 = 0.828.
results/metrics/forecast_metrics_per_cluster.csv	One row per cluster (-3 to 3): n_households, n_rows, mae, rmse, mape, r2.
results/metrics/feature_importance_global.csv	14 rows: feature name + split-count importance from global model.
results/figures/feature_importance_global.png	Horizontal bar chart of top 14 features.
results/figures/actual_vs_predicted_clusters.png	2×2 grid: mean actual vs predicted per cluster across 366 days of 2024.
results/figures/residuals_per_cluster.png	1×4 residual histograms per cluster.
results/models/lgbm_cluster_0.pkl	Trained LightGBM model for cluster 0.
results/models/lgbm_cluster_1.pkl	Trained LightGBM model for cluster 1.
results/models/lgbm_cluster_2.pkl	Trained LightGBM model for cluster 2.
results/models/lgbm_cluster_3.pkl	Trained LightGBM model for cluster 3.
results/models/lgbm_global.pkl	Global model trained on all regular households. Used as dataset-level baseline.

05_Probabilistic_Forecasting.ipynb - Prediction Intervals

Extends the cluster-based forecasts with calibrated 80% prediction intervals using quantile regression. Instead of a single predicted value, each household-day gets three values: lower (q=0.10), predicted (q=0.50), and upper (q=0.90).

Why it was added:

Cluster 1 (strong winter peak) has R^2=0.884 but MAE=3.149 kWh. The model tracks the shape of winter peaks correctly but underestimates their magnitude. Wide prediction intervals in December–February communicate this uncertainty honestly rather than hiding it behind a single point estimate. Cluster 3 (summer-peaking) benefits similarly because summer spike amplitude varies year to year.

What it does:

- Uses the same feature engineering pipeline as NB04 (identical lag features, rolling windows, calendar features, household mean). No new packages required.
- Trains 3 LightGBM models per cluster using objective='quantile'. All other hyperparameters identical to NB04. Total: 12 cluster quantile models + 3 global fallback models = 15 models.
- Quantile 0.10 = lower bound. Quantile 0.50 = median point estimate. Quantile 0.90 = upper bound. Pinball loss is minimised at each quantile.
- Enforces monotonicity after prediction: lower ≤ median ≤ upper. All values clipped to ≥ 0.
- Degenerate households: interval is ±1 MAD (median absolute deviation of non-zero 2023 values). MAD is used instead of std because it is robust when many values are zero.
- Evaluates: coverage (fraction of actuals inside interval, target = 80%), PIAW (mean interval width in kWh/day), Winkler score (combines coverage penalty + width, lower is better), miscoverage direction.

Results:

Cluster	Name	N HH	MAE (q50)	Coverage	PIAW kWh/day	Winkler
-3	zero_heavy	501	4.404	30.4%	4.797	28.61
-2	near_constant	398	0.335	31.1%	0.274	2.52
-1	all_zero	154	0.080	99.1%	0.000	0.80
0	moderate_winter_leaning	6,359	1.954	79.1%	6.017	9.22
1	strong_winter_peaking	2,757	3.143	75.1%	8.526	15.05
2	lower_flatter_weekend_lift	5,807	1.954	79.3%	5.941	9.25
3	summer_peaking	1,571	2.144	78.5%	6.383	10.39
ALL	overall	17,547	2.174	76.2%	6.201	10.58

Cluster 1 has the widest intervals (PIAW = 8.53 kWh/day) concentrated in November through February. This is the correct behaviour - the model is more confident in summer when consumption is low and stable, and widens the interval in winter when spike magnitude is uncertain. The household-level coverage distribution has mean 76.2% with std 13.6%, meaning most households are near the 80% target but 844 households have coverage below 60%.

Files saved:

File	Contents
results/predictions/predictions_2024_probabilistic.csv	6,422,202 rows. Columns: household_id, date, lower, predicted, upper, actual, cluster, cluster_name, model_used, interval_width.
results/metrics/forecast_metrics_probabilistic.csv	One row per cluster: point_mae, point_r2, coverage, piaw, winkler_score, miscoverage_low, miscoverage_high.
results/metrics/household_coverage.csv	One row per household (17,547 rows): household_id, cluster, coverage, piaw, mae.
results/figures/prob_intervals_per_cluster.png	2×2 grid: mean actual vs median predicted with shaded 80% PI per cluster across 366 days.
results/figures/prob_cluster1_winter_focus.png	Cluster 1 full year with winter months highlighted in orange. Annotation showing peak interval width.
results/figures/prob_interval_width_distribution.png	Box plots of per-day interval width by cluster + seasonal width curve for clusters 1 and 3.
results/figures/prob_household_coverage.png	Coverage histogram for all 17,547 households. Shows distribution around the 80% target.
results/figures/prob_calibration_scatter.png	Coverage vs PIAW scatter, one bubble per cluster. Bubble size proportional to n_households.
results/figures/prob_example_household_c1.png	Single cluster-1 household (ID 151979): full 2024 actual vs interval trace. Coverage 75.1%, PIAW 8.21 kWh, MAE 2.90 kWh/day.
results/models/probabilistic/lgbm_cluster0_q10.pkl	q=0.10 model for cluster 0. (Similarly named files exist for q50, q90, and clusters 1–3.)
results/models/probabilistic/lgbm_global_q10.pkl	Global fallback quantile models (q10, q50, q90).

5. Source files (src/)

utils/config.py

Single source of truth for all constants. Every notebook imports from here. Key values:

Constant	Value	Purpose
RANDOM_SEED	42	Reproducibility across all random operations
FINAL_K	4	Number of clusters in the frozen final clustering
LAG_FEATURES	[1, 7, 14, 30]	Lag days used as forecasting features
ROLLING_WINDOWS	[7, 14, 30]	Rolling window sizes for mean and std features
FORECAST_HORIZON	366	Days to predict in 2024
DIAGNOSTICS_DIR	results/diagnostics/	Where cluster assignment files live
PREDICTIONS_DIR	results/predictions/	Where prediction CSVs are saved
METRICS_DIR	results/metrics/	Where metric CSVs are saved
FIGURES_DIR	results/figures/	Where plots are saved

utils/data_loader.py

Two functions: load_train_data() returns sample_23.csv as a wide DataFrame (17547 × 365) with household ID as index. load_test_data() returns sample_24.csv (17547 × 366). Both called at the start of every notebook.

preprocessing/data_cleaning.py

DataCleaner class. The method prepare_for_feature_clustering() separates degenerate households using three rules: all_zero if zero_pct == 1.0, near_constant if std < 0.5 kWh, zero_heavy if zero_pct > 0.30. Outliers among regular households are kept deliberately.

preprocessing/eda_analysis.py

All analytical functions used in NB01 and NB03: compute_household_statistics(), build_phase1_clustering_summary(), split_households_for_clustering(), aggregate_by_day_of_week(), aggregate_by_month(), z_normalize_rows(), categorize_households(), detect_normalized_trend().

preprocessing/feature_engineering.py

FeatureEngineer class. extract_main_clustering_features() produces the (16494, 32) feature matrix. Features are organized in the 5 blocks described in the NB02 section. The 3.25× weight on shape features is the most important methodological decision - it ensures clustering groups households by how consumption rises and falls through the year, not just by how much they consume.

clustering/clustering.py

HouseholdClusterer class. run_feature_ward_clustering(train_data, k) runs the full pipeline and returns cluster labels. prepare_clustering_handoff(train_data) runs the final frozen k=4 clustering and saves the four CSV files to results/diagnostics/. Once called, the clustering is frozen and should not be re-run.

clustering/clustering_evaluation.py

ClusteringEvaluator class. rank_k_candidates() scores candidate clusterings by composite score (silhouette + balance + minimum size). shortlist_reasonable_k() filters out solutions with very small or very large clusters. select_best_k() picks the top candidate. This evaluator suggested k=3; the final choice of k=4 was a manual override documented in NB03.

clustering/cluster_reporting.py

ClusterReporter class. build_report_bundle() generates diagnostic tables and profile summaries per cluster. mean_monthly_profile_by_cluster() and mean_weekly_profile_by_cluster() return the DataFrames that become cluster_monthly_profiles.csv and cluster_weekly_profiles.csv.

6. All output files - complete list

Everything in results/ that exists after all five notebooks have been run.

results/diagnostics/

File	Shape / Contents
cluster_assignments.csv	17,547 rows. Index = household_id. Column = cluster (integer -3 to 3).
cluster_sizes.csv	4 rows × 3 cols (cluster label, name, count). Regular clusters only.
cluster_monthly_profiles.csv	4 rows × 13 cols. Mean monthly consumption index per regular cluster.
cluster_weekly_profiles.csv	4 rows × 8 cols. Mean day-of-week consumption index per regular cluster.

results/metrics/

File	Contents
forecast_metrics_overall.csv	1 row: mae=2.1749, rmse=4.9824, mape=188.9, r2=0.828 for all 17,547 households.
forecast_metrics_per_cluster.csv	7 rows (-3 to 3): cluster, cluster_name, n_households, n_rows, mae, rmse, mape, r2.
feature_importance_global.csv	14 rows: feature name + split-count importance from global model. Top 3: lag_1 (3882), day_of_year (3484), household_mean (2842).
forecast_metrics_probabilistic.csv	7 rows (-3 to 3): point_mae, point_r2, coverage, piaw, winkler_score, miscoverage_low, miscoverage_high.

File	Contents
household_coverage.csv	17,547 rows: household_id, cluster, coverage, piaw, mae. Per-household PI calibration stats.

results/predictions/

File	Contents
predictions_2024.csv	6,422,202 rows. Columns: household_id, date, predicted, actual, cluster, cluster_name, model_used. Point forecasts from cluster LightGBM models (regular HH) and historical mean (degenerate HH).
predictions_2024_probabilistic.csv	6,422,202 rows. Columns: household_id, date, lower, predicted, upper, actual, cluster, cluster_name, model_used, interval_width. Probabilistic forecasts with 80% prediction intervals.

results/models/

File	Contents
lgbm_cluster_0.pkl	LightGBM point model for cluster 0 (moderate_winter_leaning). Trained on 2.13M rows.
lgbm_cluster_1.pkl	LightGBM point model for cluster 1 (strong_winter_peaking). Trained on 923K rows.
lgbm_cluster_2.pkl	LightGBM point model for cluster 2 (lower_flatter_weekend_lift). Trained on 1.95M rows.
lgbm_cluster_3.pkl	LightGBM point model for cluster 3 (summer_peaking). Trained on 526K rows.
lgbm_global.pkl	Global LightGBM model trained on all 5.53M regular-household rows. Used as dataset-level baseline.
probabilistic/lgbm_cluster0_q10.pkl	Quantile 0.10 model for cluster 0. Lower bound of 80% PI.
probabilistic/lgbm_cluster0_q50.pkl	Quantile 0.50 (median) model for cluster 0. Point estimate.
probabilistic/lgbm_cluster0_q90.pkl	Quantile 0.90 model for cluster 0. Upper bound of 80% PI.
probabilistic/lgbm_cluster1_q*.pkl	Same three quantile models for cluster 1 (widest intervals - winter peak uncertainty).
probabilistic/lgbm_cluster2_q*.pkl	Same three quantile models for cluster 2.
probabilistic/lgbm_cluster3_q*.pkl	Same three quantile models for cluster 3.
probabilistic/lgbm_global_q10/q50/q90.pkl	Global fallback quantile models for unmatched households.

results/figures/

File	What it shows
feature_importance_global.png	Top 14 features by split count (global model). lag_1 and day_of_year dominate.
actual_vs_predicted_clusters.png	2×2 grid: mean actual vs predicted per cluster, all 366 days. Cluster 1 visibly underpredicts Jan–Feb peak.
residuals_per_cluster.png	1×4 residual histograms. Cluster 1 shows positive skew (underprediction). Cluster -3 has wide tails.
prob_intervals_per_cluster.png	2×2 grid: actual vs median predicted with shaded 80% PI per cluster. Cluster 1's wide band is clearly visible.
prob_cluster1_winter_focus.png	Cluster 1 full year. Winter months highlighted in orange. Annotation showing peak interval width.
prob_interval_width_distribution.pn g	Box plots of interval width by cluster + seasonal width curve for clusters 1 and 3 by month.
prob_household_coverage.png	Coverage histogram for all 17,547 households. Mean = 76.2%, target = 80%.
prob_calibration_scatter.png	Coverage vs mean interval width scatter. One bubble per cluster, size = n_households.
prob_example_household_c1.png	Household 151979 (cluster 1): full 2024 actual vs interval. Coverage 75.1%, PIAW 8.21 kWh, MAE 2.90 kWh/day.

7. Key numbers

Fact	Value
Total households	17,547
Training period	2023-01-01 to 2023-12-31 (365 days)
Test period	2024-01-01 to 2024-12-31 (366 days, leap year)
Regular households (clustered)	16,494
Degenerate households	1,053 (6% of total)
Final number of clusters	4 (algorithm suggested 3, overridden to 4)
Cluster 0 size	6,359 HH (38.6% of regular)
Cluster 1 size	2,757 HH (16.7%)
Cluster 2 size	5,807 HH (35.2%)
Cluster 3 size	1,571 HH (9.5%)
Features for clustering	32 (5 blocks, shape-normalized at 3.25× weight)
Features for forecasting	16 (4 lags, 6 rolling, 5 calendar, 1 household mean)
Overall cluster model MAE	2.1749 kWh/day
Overall cluster model R^2	0.828
Best cluster MAE	1.953 kWh/day (clusters 0 and 2)

Fact	Value
Worst regular cluster MAE	3.149 kWh/day (cluster 1, strong winter peak)
Best cluster R^2	0.885 (cluster 3, summer-peaking)
Global model MAE	Run evaluate_global.py to compute - not yet compared
PI nominal coverage target	80% (q0.10 to q0.90)
PI actual coverage achieved	76.2%
Cluster 1 mean interval width	8.53 kWh/day (widest - winter peak uncertainty)
Cluster 2 mean interval width	5.94 kWh/day (narrowest - flat profile, most certain)
Top feature by importance	lag_1 (3882) - yesterday's consumption
Second feature	day_of_year (3484) - seasonal position
Third feature	household_mean (2842) - long-run average